

**simon-5507-05-slides**

# Topics to be covered

- What you will learn
  - Labels and formats
  - Descriptive statistics
  - Correlations and scatterplots
  - Boxplots and group statistics
  - Investigate anomaly

# Overview

- Mix of categorical and continuous variables
  - Review methods for continuous variables
  - Review methods for categorical variables
  - New methods for mix of categorical and continuous variables

## Speaker notes

Today, you will analyze some data sets that have a mix of categorical and continuous variables. The first data set looks at pulmonary function in a group of children.

You can find a description of this data set at

<http://jse.amstat.org/datasets/fev.txt>

# Labels and formats

- Document your data
  - Formats for number codes
  - Labels for all variables

## Speaker notes

Documenting your code is important. If your categorical data uses number codes, make sure to define what those codes represent using proc format.

Also use labels for all variables, with the possible exception of simple variables that are self-explanatory with the variable name alone.

# Documentation header (1)

```
* simon-5507-05-demo.sas
  author: Steve Simon
  date created: 2018-11-27
  purpose: to illustrate how to work with
           data that has a mix of categorical and
           continuous variables.
  license: public domain;
```

Speaker notes

- Comments on the code: Documentation header;



# File locations (2)

```
%let path=q:/5507-2025b/05;
```

```
ods pdf  
  file("&path/results/simon-5507-05-demo.pdf");
```

```
filename raw_data  
  "&path/data/fev.txt";
```

```
libname perm  
  "&path/data";
```

Speaker notes

- Comments on the code: File locations;

# Reading the data using a data step (3)

```
data perm.fev;  
  infile raw_data delimiter=",";  
  input  
    age 2-3  
    fev 5-11  
    ht 12-16  
    sex 19  
    smoke 25;  
  label  
    age=Age in years  
    fev=Forced Expiratory Volume (liters)  
    ht=Height in inches  
    sex=Sex  
    smoke=Smoking status
```

## Speaker notes

- Comments on the code: Reading the data using a data step

The data file is comma delimited and the first row includes variable names.

Normally, this means that you can save a bit of time by using proc import, but I chose to read in the data using a data step. The number of variables was so small that this didn't matter that much. It also allowed me to define variable labels in the initial data step rather than later.;

# Label your categorical variables (4)

```
proc format;  
  value fsex  
    0 = "Female"  
    1 = "Male"  
  ;  
  value fsmoke  
    0 = "Nonsmoker"  
    1 = "Smoker"  
  ;  
run;
```

## Speaker notes

- Comments on the code: Label your categorical variables;

# Print the first ten rows of data (5)

```
proc print
  data=perm.fev(obs=10);
  format
    sex fsex.
    smoke fsmoke.;
  title1 "Pulmonary function study";
  title2 "There are no obvious problems with this dataset";
  footnote1 "Created by Steve Simon on &sysdate using SAS &sysver";
run;
```

## Speaker notes

- Comments on the code: Print the first ten rows of data

It's always a good idea to peek at the first few rows of data.



# Live demo, part 1 (review)

- documentation header
- file locations
- data step
- label subcommand
- proc format
- format subcommand

# Break #1

- What you have learned
  - Labels and formats
- What's coming next
  - Descriptive statistics

# Descriptive statistics

- Use proc freq to look for
  - Inconsistencies
  - Rare/missing category levels
- Use proc means to look for
  - Unusual minimum/maximum values
  - Zero standard deviation

## Speaker notes

There is a mix of categorical and continuous variables in this data set. Recall that you use `proc freq` for categorical variables and `proc means` for continuous variables.

Always get in the habit of checking for missing values.

Look for problems. This could mean a lot more categories than you expected, a particular category level that is unexpectedly small, or multiple categories caused by misspelling or inconsistent capitalization. There are no problems here.

Look for minimum or maximum values that are unusual. Also make sure that you don't have a continuous variable that is constant (zero variation).

# Get statistics for categorical variables (6)

```
proc freq  
    data=perm.fev;  
    tables sex smoke / missing;  
    format  
        sex fsex.  
        smoke fsmoke.;  
    title2 "Frequency counts";  
run;
```

## Speaker notes

- Comments on the code: Get statistics for categorical variables;

# Get statistics for continuous variables (7)

```
proc means  
    n nmiss mean std min max  
    data=perm.fev;  
    var age fev ht;  
    title2 "Descriptive statistics";  
run;
```

## Speaker notes

- Comments on the code: Get statistics for continuous variables;



# Live demo, part 2 (review)

- proc freq
- proc means

# Break #2

- What you have learned
  - Descriptive statistics
- What's coming next
  - Correlations and scatterplots

# Look at relationships

- Between two continuous variables
  - Correlations, Scatterplots
- Between two categorical variables
  - Crosstabulations
- Between a categorical and a continuous variables
  - Boxplots

## Speaker notes

After looking at descriptive statistics for individual variables, you should look for relationships between pairs of variables.

Correlations, covered earlier, are a simple way to examine relationships between two continuous variables.

Remember the cut-offs. A correlation between +0.7 and 1.0 implies a strong positive association. A correlation between +0.3 and +0.7 implies a weak positive association. A correlation between -0.3 and +0.3 implies little or no association. A correlation between -0.3 and -0.7 implies a weak negative association. A correlation between -0.7 and -1.0 implies a strong negative association.;

# Compute a correlation matrix (8)

```
proc corr  
    data=perm.fev  
    noprint  
    outp=correlations;  
    var age fev ht;  
run;
```

## Speaker notes

- Comments on the code: Compute a correlation matrix;

# Round the correlations (9)

```
data correlations;  
  set correlations;  
  if _type_ NE "CORR" then delete;  
  drop _type_;  
  age=round(age, 0.01);  
  fev=round(fev, 0.01);  
  ht=round(ht, 0.01);  
run;
```

Speaker notes

- Comments on the code: Round the correlations;



# Print the correlations (10)

```
proc print  
    data=correlations;  
    title2 "All variables show a positive correlations";  
run;
```

## Speaker notes

- Comments on the code: Print the correlations

With a small number of variables, there is no need to sort the correlations when there are just a few of them.;

# Draw scatterplots (11)

```
ods graphics /  
    height=6 in width=6 in;  
  
proc sgplot  
    data=perm.fev;  
    pbspline x=age y=fev /  
        markerattrs=(size=10 symbol=circle color=black)  
        lineattrs=(pattern=dash color=red);  
    title2 "There is a positive association, with slight bends at low and high ages";  
run;  
  
ods graphics on / reset=all;
```

## Speaker notes

- Comments on the code: Draw scatterplots;

# Live demo, part 3 (review)

- proc corr
- proc sgplot
  - scatter subcommand
    - markerattrs option
  - pbspline subcommand
    - lineattrs option

# Break #3

- What you have learned
  - Correlations and scatterplots
- What's coming next
  - Boxplots and group statistics

# Boxplots

- Summarizes five numbers
  - Lower whisker: minimum value
  - Bottom of box: 25th percentile
  - Middle of box: 50th percentile
  - Top of box: 75th percentile
  - Upper whisker: maximum value
- Options
  - Highlight individual outliers
  - Plus sign (mean)

## Speaker notes

When you want to look at a relationship between a categorical variable and a continuous variable, you should use a boxplot.

The bottom and top of the box represents the 25th and 75th percentiles, respectively. A thin line, or whisker, is drawn down to the minimum value and up to the maximum value. Extreme values are shown as individual data points.

In the fev data set, as you will see in a minute, there is an odds result involving fev. Smokers seem to have a much higher Fev than non-smokers. This is quite surprising.



# Group statistics

- proc means with a by subcommand
  - produces statistics for each subgroup
- Look for
  - Shifts in the mean
  - Changes in variation

## Speaker notes

the `by` subcommand in `proc means` will produce statistics on a continuous variable, but separately for each subgroup identified by a level of your categorical variable.

Look for shifts in the mean and changes in variation. Don't overreact to small changes in variation. Typically, you need to see at least a doubling or tripling of the standard deviation from one level to another before worrying about it.

# Draw a boxplot (12)

```
ods graphics /  
    height=2.5 in width=6 in;  
  
proc sgplot  
    data=perm.fev;  
    hbox fev / category=smoke nofill;  
    yaxis label=" ";  
    format smoke fsmoke.;  
    title2 "Smokers tend to have higher fev values";  
    title3 "This is a surprising and counter-intuitive finding";  
run;  
  
ods graphics on / reset=all;
```

## Speaker notes

- Comments on the code: Draw a boxplot;

# Compute statistics with a by statement (13)

```
proc sort  
    data=perm.fev;  
    by smoke;  
run;
```

```
proc means  
    n nmiss mean std min max  
    data=perm.fev;  
    var fev;  
    by smoke;  
    title2 "Descriptive statistics by group";  
run;
```

## Speaker notes

- Comments on the code: Compute statistics with a by statement;

# Live demo, part 4

- `proc sgplot`
  - `hbox` subcommand
    - `category` option
    - `nofill` option

# Break #4

- What you have learned
  - Boxplots and group statistics
- What's coming next
  - Investigate anomaly



# Investigating potential confounders

- Could a third variable account for the odd result?
- Visualize three variables at once
  - Use panels
  - Use colors
  - Use markers
  - Use linetypes
  - Use size

## Speaker notes

The unexpected finding in the SAS program, children who smoke have better lung function than children who don't. There are at least two theories.

First, maybe smoking is mostly limited to boys and boys tend to have larger lung capacities than girls.

Second, maybe smoking is related age. Only older children smoke and older children tend to have larger lung capacities.

This is a classic problem in Statistics. Could an unexpected relationship between two variables be caused by their relationships to a third variable, a potential confounding variable.

To investigate this, you need a way to incorporate that third variable into your visualization. There are multiple ways to do this. Some work well for potential confounders that are continuous, some work well for potential confounders that are categorical, and some work well for either type.

A common approach is to use panels. Each panel represents a level of your categorical confounder or it represents a limited range of your continuous confounder.

You can use different discrete colors to represent different values of your categorical confounder or a range of colors to represent values of your continuous confounder.

In a scatter plot, you can use different markers for different category levels. This could be circles for one level and plus signs for the other level.

You can use different linetypes for your trend line or smoothing spline: solid versus dashed versus dotted.

You can use the size of a marker to represent different values of your continuous confounder.

You can also use some combination of these.

# Boxplots in separate panels (14)

```
ods graphics /  
    height=3.5 in width=6 in;  
  
proc sgpanel  
    data=perm.fev;  
    panelby sex / columns=1 rows=2;  
    hbox fev / category=smoke nofill;  
    format smoke fsmoke. sex fsex.;  
    title2 "Our first guess is that sex is a confounder";  
    title3 "This is not supported by the data";  
run;  
  
ods graphics on / reset=all;
```

## Speaker notes

- Comments on the code: Boxplots in separate panels

The group option in hbox draws separate boxplots for each combination of group and category.;

# Important setup options for grouped plots (15)

```
ods graphics /  
  height=6 in width=6 in  
  attrpriority=none;
```

```
proc sort  
  data=perm.fev;  
  by smoke;  
run;
```

## Speaker notes

- Comments on the code: Important setup options for grouped plots

The `attrpriority` option tells SAS how to cycle through various attributes (color, maker, linestyle) to distinguish among different groups. By default, SAS will cycle through a list of colors and only switch to other features after that list is exhausted. Most of the time, you do not want this option, so specify `none`.

It is important to sort your data when you have groups, to control how colors, markers, and linestyle are assigned to each group.;

# Distinguishing groups in a scatterplot (16)

```
proc sgplot
    data=perm.fev;
    scatter x=age y=fev /
        group=smoke
        markerattrs=(size=10);
    format smoke fsmoke.;
    styleattrs
        datacontrastcolors=(lightgray black)
        datasymbols=(circle plus);
    title2 "Second hypothesis is that age is a confounder";
    title3 "This hypothesis is supported by the data";
run;

ods graphics on / reset=all;
```

## Speaker notes

- Comments on the code: Distinguishing groups in a scatterplot

The `group` option tells SAS to make distinctions in a graph using a categorical variable.

The `styleattrs` statement tells SAS what particular colors, symbols, linestyles, etc. to use to make group distinctions.;



# Live demo, part 5

- `proc sgpanel`
  - `panelby` subcommand
- `proc sgplot`
  - `styleattrs` subcommand
  - `datasymbols`
  - `datacontrastcolors`

# Summary

- What you have learned
  - Labels and formats
  - Descriptive statistics
  - Correlations and scatterplots
  - Boxplots and group statistics
  - Investigate anomaly